

< Previous	 	 					Next >
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1.2.3 The vector 2-norm (Euclidean length)

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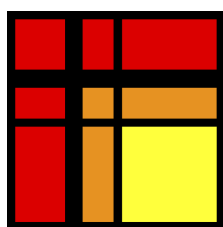
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1/1 point (graded)



Read Unit 1.2.3 of the notes. [[LINK](#)].

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🗨️ Proving vector 2-norm is a norm proof clarification

4

	Hello, When proving the triangle inequality holds in the 2-norm is a norm proof, the following step was made: $(x+y)^H * (x+y) = x^H * x + y^H * x...$	
?	Additional typos in 1.2.3.2. and 1.2.3.3? In the discussion in Remark 1.2.3.2. below the triangle concerning the "...component of x in the direction of a vector y of unit length...", what is t...	2
?	Typo in Remark 1.2.3.2. of the notes?	3

Homework 1.2.3.2

1.0/1.0 point (graded)
Prove that the vector 2-norm is a norm.

☒ done



Solution:

For a complete solution see the [notes](#).

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Homework 1.2.3.3

1.0/1.0 point (graded)
Partition $x \in \mathbb{C}^m$ into subvectors:

$$x = \begin{pmatrix} x_0 \\ x_1 \\ \vdots \\ x_{M-1} \end{pmatrix}.$$

$\|x_i\|_2 \leq \|x\|_2$

ALWAYS

Answer: ALWAYS

Solution:

For a complete solution see the [notes](#).

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